AI Engineer Case

Document Question Answering -- Coding Assignment

Background

In this case, you will build an AI-based system that can answer questions based on a provided set of documents.

You will be given:

- a set of documents,
- a set of questions that the system should be able to answer,
- access to a language model hosted in Azure.

The information required to answer the questions is contained in the provided documents and cannot be assumed to be known by the language model beforehand.

Task

Build a solution that takes a question as input and returns the best possible answer based on the information available in the provided documents.

How you choose to solve the problem is up to you.

We are primarily interested in how you approach the problem, the technical decisions you make, and how well the resulting solution works.

Some answers require combining or calculating numbers found across multiple documents (for example tables of prices, tariffs, or recipe quantities), not just retrieving a single passage of text.

Technical Setup

The solution should be able to run locally on a standard development computer.

The language model is provided as an external service through Azure. The endpoint, deployment name, API version, and API key you need are provided in the accompanying file LLM_instructions.txt, delivered separately alongside this document.

You are otherwise free to use the technologies, libraries, and tools of your choice, as long as the solution can be run locally according to the instructions in the repository.

A graphical user interface is not required.

The solution should be runnable from the command line by passing a question as an argument, for example:

python agent.py "How much sugar is needed for 1,000 buns?"

It is fine to run a separate one-time preprocessing step before this (for example a script named **preprocessing.py**) if your solution needs to convert or prepare the documents in some way before they can be queried. How you choose to do this, if at all, is up to you -- it does not need to be described here beyond making clear in your instructions whether such a step is required and how to run it.

Requirements

The solution should:

- run locally,
- be runnable from the command line with a question as input (and an optional preprocessing step, if used),
- be able to answer the questions included in the case,
- base its answers on information contained in the provided documents,
- indicate the source material used to support an answer,
- handle situations where the available information is insufficient to provide a confident answer,
- handle situations where the available information is ambiguous or conflicting, and be clear about how that was resolved.

Delivery

The solution should be placed in a Git repository and shared with us before the follow-up interview.

The repository should contain:

- the source code,
- instructions for installing and running the solution,
- a brief description of the solution and the key technical decisions you made.

Do not include credentials, API keys, or other secrets in the repository. Keep the contents of LLM_instructions.txt out of version control (for example via .gitignore).

Time

We recommend spending approximately 3 hours on the case.

The solution does not need to be production-ready or include everything you would build in a real-world system. We are more interested in the decisions and trade-offs you make within the available time.

You are welcome to use AI assistants and other development tools in the same way you would in your day-to-day work.

Evaluation

After reviewing your solution, we will also test it against a small number of additional questions of the same kind, covering the same set of documents. These questions are not included in this case, so your solution should generalize rather than be tailored only to the questions provided.

Follow-up

During the follow-up interview, we will review the solution together. We will discuss topics such as:

- how you approached the problem,

- the technical decisions you made,
- the trade-offs you considered,
- how the solution handles different types of questions and source material,
- what you would change or develop further for a production environment.

You may also be asked to run the solution against additional questions or scenarios during the interview.

Provided files: candidate documents (01-11), the questions document (11 - Questions.pdf), and LLM_instructions.txt (LLM access credentials, delivered separately).